Dongruo Zhou

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Professional Experience

Assistant Professor. *Department of Computer Science, Indiana University Bloomington* Aug 2023 -

Education

Ph.D. in Computer Science. *University of California, Los Angeles. Advisor: Prof. Quanquan Gu* Sept 2018 - June 2023
Ph.D. student in Systems and Information Engineering. *University of Virginia. Advisor: Prof. Quanquan Gu* Aug 2017 - Aug 2018
B.Sc in Pure and Applied Mathematics. *Tsinghua University* Aug 2013 - Jul 2017

Research interest

My research focuses on uniting **reinforcement learning**, **machine learning**, and **optimization** to design adaptive and efficient **autonomous agents**. Through this integration, I aim to advance agents that can navigate complex, dynamic environments with robust decision-making capabilities.

teaching experience

Instructor at Indiana University Bloomington
B455, Principle on Machine Learning Jan 2025 - May 2025
B565, Data Mining Aug 2024 - Dec 2024
B659, Offline Decision Making Jan 2024 - May 2024
B565, Data Mining Aug 2023 - Dec 2023
Teaching Assistant at University of California, Los Angeles
CS 161, Fundamentals of AI Jan 2020 - Dec 2020
Teaching Assistant at University of Virginia
APMA 3100, Probability Feb 2018 – May 2018
DS 6002, Introduction to Data Science Aug 2017 – Feb 2018

Publication (* indicates equal contribution)

Full publication list: https://scholar.google.com/citations?user=1780wr0AAAAJhl=en
Conference Publications

42. Federated In-Context Learning: Iterative Refinement for Improved Answer Quality,
Ruhan Wang* , Zhiyong Wang*, Chengkai Huang*, Rui Wang, Tong Yu, Lina Yao, John C.S. Lui, Dongruo Zhou,
International Conference on Machine Learning (ICML), 2025.
41. Provable Zero-Shot Generalization in Offline Reinforcement Learning,
Zhiyong Wang , Chen Yang, John C.S. Lui, Dongruo Zhou,
International Conference on Machine Learning (ICML), 2025.
40. Sample and Computationally Efficient Continuous-Time Reinforcement Learning with General Function Approximation,
Runze Zhao*, Yue Yu*, Yiyue Adams Zhu, Chen Yang, Dongruo Zhou,
Uncertainty in Artificial Intelligence (UAI), 2025.
39. Safe Decision Transformer with Learning-based Constraints,
Ruhan Wang, Dongruo Zhou,
Learning for DynamIcs Control (L4DC), 2025.
38. Breaking the $\log(1/\Delta_2)$ Barrier: Better Batched Best Arm Identification with Adaptive Grids,
Tianyuan Jin*, Qin Zhang*, Dongruo Zhou*,
International Conference on Learning Representations (ICLR), 2025.
37. Model-based RL as a Minimalist Approach to Horizon-Free and Second-Order Bounds,
Zhiyong Wang, Dongruo Zhou, John C.S. Lui, Wen Sun,
International Conference on Learning Representations (ICLR), 2025.
36. Variance-Dependent Regret Bounds for Non-stationary Linear Bandits,
Zhiyong Wang, Jize Xie, Yi Chen, John C.S. Lui, Dongruo Zhou,
International Conference on Artificial Intelligence and Statistics (AISTATS), 2025.
35. Uncertainty-Aware Reward-Free Exploration with General Function Approximation,
Junkai Zhang*, Weitong Zhang*, Dongruo Zhou, Quanquan Gu,
International Conference on Machine Learning (ICML), 2024.
34. Risk Bounds of Accelerated SGD for Overparameterized Linear Regression,
Xuheng Li, Yihe Deng, Jingfeng Wu, Dongruo Zhou, Quanquan Gu,
International Conference on Learning Representations (ICLR), 2024.

33. Provably efficient representation selection in Low-rank Markov Decision Processes: from online to offline RL,
Weitong Zhang, Jiafan He, Dongruo Zhou, Amy Zhang, Quanquan Gu,
Uncertainty in Artificial Intelligence (UAI), 2023.
32. Variance-Dependent Regret Bounds for Linear Bandits and Reinforcement Learning: Adaptivity and Computational Efficiency,
Heyang Zhao, Jiafan He, Dongruo Zhou, Tong Zhang, Quanquan Gu,
Conference on Learning Theory (COLT), 2023.
31. Nearly Minimax Optimal Regret for Learning Linear Mixture Stochastic Shortest Path,
Qiwei Di, Jiafan He, Dongruo Zhou, Quanquan Gu,
International Conference on Machine Learning (ICML), 2023.
30. Nearly Minimax Optimal Reinforcement Learning for Linear Markov Decision Processes,
Jiafan He, Heyang Zhao, Dongruo Zhou, Quanquan Gu,
International Conference on Machine Learning (ICML), 2023.
29. Optimal Online Generalized Linear Regression with Stochastic Noise and Its Application to Heteroscedastic Bandits,
Heyang Zhao, Dongruo Zhou, Jiafan He, Quanquan Gu,
International Conference on Machine Learning (ICML), 2023.
28. Nearly Optimal Algorithms for Linear Contextual Bandits with Adversarial Corruptions,
Jiafan He, Dongruo Zhou, Tong Zhang, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2022.
27. Computationally Efficient Horizon-Free Reinforcement Learning for Linear Mixture MDPs,
Dongruo Zhou, Quanquan Gu
Advances in Neural Information Processing Systems (NeurIPS), 2022; Oral. [Code]
26. Learning Two-Player Markov Games: Neural Function Approximation and Correlated Equilibrium,
Chris Junchi Li*, Dongruo Zhou*, Quanquan Gu, Michael I. Jordan
Advances in Neural Information Processing Systems (NeurIPS), 2022.
25. Dimension-free Complexity Bounds for High-order Nonconvex Finite-sum Optimization,
Dongruo Zhou, Quanquan Gu,
International Conference on Machine Learning (ICML), 2022.
24. Learning Neural Contextual Bandits through Perturbed Rewards,
Yiling Jia, Weitong Zhang, Dongruo Zhou, Quanquan Gu, Hongning Wang,
International Conference on Learning Representations (ICLR), 2022.
23. Nearly Minimax Optimal Regret for Learning Infinite-horizon Average-reward MDPs with Linear Function Approximation,
Yue Wu, Dongruo Zhou, Quanquan Gu,
International Conference on Artificial Intelligence and Statistics (AISTATS), 2022.
22. Near-optimal Policy Optimization Algorithms for Learning Adversarial Linear Mixture MDPs,
Jiafan He, Dongruo Zhou, Quanquan Gu,
International Conference on Artificial Intelligence and Statistics (AISTATS), 2022.
21. Faster Perturbed Stochastic Gradient Methods for Finding Local Minima,
Zixiang Chen*, Dongruo Zhou*, Quanquan Gu,
Algorithmic Learning Theory (ALT), 2022.
20. Almost Optimal Algorithms for Two-player Markov Games with Linear Function Approximation,
Zixiang Chen, Dongruo Zhou, Quanquan Gu,
Algorithmic Learning Theory (ALT), 2022.
19. Uniform-PAC Bounds for Reinforcement Learning with Linear Function Approximation,
Jiafan He, Dongruo Zhou, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2021.
18. Pure Exploration in Kernel and Neural Bandits,
Yinglun Zhu*, Dongruo Zhou*, Ruoxi Jiang*, Quanquan Gu, Rebecca Willett, Robert D. Nowak,
Advances in Neural Information Processing Systems (NeurIPS), 2021. [Code]
17. Variance-Aware Off-Policy Evaluation with Linear Function Approximation,
Yifei Min*, Weitong Zhang*, Dongruo Zhou, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2021.
16. Provably Efficient Reinforcement Learning with Linear Function Approximation under Adaptivity Constraints,
Tianhao Wang*, Dongruo Zhou*, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2021.
15. Reward-Free Model-Based Reinforcement Learning with Linear Function Approximation,
Weitong Zhang, Dongruo Zhou, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2021.
14. Iterative Teacher-Aware Learning,
Luyao Yuan, Dongruo Zhou, Junhong Shen, Jingdong Gao, Jeffrey L Chen, Quanquan Gu, Ying Nian Wu, Song-Chun Zhu,
Advances in Neural Information Processing Systems (NeurIPS), 2021.

13. Nearly Minimax Optimal Reinforcement Learning for Discounted MDPs,
Jiafan He, Dongruo Zhou, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2021.
12. Nearly Minimax Optimal Reinforcement Learning for Linear Mixture Markov Decision Processes,
Dongruo Zhou, Quanquan Gu, Csaba Szepesvári,
Conference on Learning Theory (COLT), 2021.
11. Logarithmic Regret for Reinforcement Learning with Linear Function Approximation,
Jiafan He, Dongruo Zhou, Quanquan Gu,
International Conference on Machine Learning (ICML), 2021.
10. Provably Efficient Reinforcement Learning for Discounted MDPs with Feature Mapping,
Dongruo Zhou, Jiafan He, Quanquan Gu,
International Conference on Machine Learning (ICML), 2021.
9. Neural Thompson Sampling,
Weitong Zhang, Dongruo Zhou, Lihong Li, Quanquan Gu,
International Conference on Learning Representations (ICLR), 2021. [Code]
8. Neural Contextual Bandits with UCB-based Exploration,
Dongruo Zhou, Lihong Li, Quanquan Gu,
International Conference on Machine Learning (ICML), 2020. [Code]
7. Closing the Generalization Gap of Adaptive Gradient Methods in Training Deep Neural Networks,
Jinghui Chen, Dongruo Zhou, Yiqi Tang, Ziyan Yang, Yuan Cao, Quanquan Gu,
International Joint Conference on Artificial Intelligence (IJCAI), 2020. [Code]
6. Accelerated Factored Gradient Descent for Low-Rank Matrix Factorization,
Dongruo Zhou, Yuan Cao, Quanquan Gu,
International Conference on Artificial Intelligence and Statistics (AISTATS), 2020.
5. Stochastic Recursive Variance-Reduced Cubic Regularization Methods,
Dongruo Zhou, Quanquan Gu,
International Conference on Artificial Intelligence and Statistics (AISTATS), 2020.
4. A Frank-Wolfe Framework for Efficient and Effective Adversarial Attacks,
Jinghui Chen, Dongruo Zhou, Jinfeng Yi, Quanquan Gu,
Association for the Advancement of Artificial Intelligence (AAAI), 2020.
3. Lower Bounds for Smooth Nonconvex Finite-Sum Optimization,
Dongruo Zhou, Quanquan Gu,
International Conference on Machine Learning (ICML), 2019.
2. Stochastic Nested Variance Reduction for Nonconvex Optimization,
Dongruo Zhou, Pan Xu, Quanquan Gu,
Advances in Neural Information Processing Systems (NeurIPS), 2018; Spotlight.
1. Stochastic Variance-Reduced Cubic Regularized Newton Methods,
Dongruo Zhou, Pan Xu, Quanquan Gu,
International Conference on Machine Learning (ICML), 2018.

Journal Publications

5. On the Convergence of Adaptive Gradient Methods for Nonconvex Optimization,
Dongruo Zhou*, Jinghui Chen*, Yuan Cao*, Ziyan Yang, Quanquan Gu.,
Transactions on Machine Learning Research (TMLR), 2024.
4. Batched Neural Bandits,
Quanquan Gu*, Amin Karbasi*, Khashayar Khosravi*, Vahab Mirrokni*, Dongruo Zhou*,
Journal of Data Science, 2024.
3. Stochastic Nested Variance Reduction for Nonconvex Optimization,
Dongruo Zhou, Pan Xu, Quanquan Gu,
Journal of Machine Learning Research (JMLR), 2020.
2. Stochastic Variance-Reduced Cubic Regularized Newton Methods,
Dongruo Zhou, Pan Xu, Quanquan Gu,
Journal of Machine Learning Research (JMLR), 2019.
1. Stochastic Gradient Descent Optimizes Over-parameterized Deep ReLU Networks,
Difan Zou*, Yuan Cao*, Dongruo Zhou, Quanquan Gu,
Machine Learning Journal (MLJ), 2019.

Workshop Publications

2. CoPS: Empowering LLM Agents with Provable Cross-Task Experience Sharing,
Chen Yang, Chenyang Zhao, Quanquan Gu, Dongruo Zhou,
Language Gamification Workshop, NeurIPS, 2024.

1. Provable Multi-Objective Reinforcement Learning with Generative Models,
Dongruo Zhou, Jiahao Chen, Quanquan Gu,
Challenges of Real-World RL (RWRL), NeurIPS, 2020.

AWARDS

Outstanding Graduate Student Research Award June 2023
Computer Science Department, University of California, Los Angeles
Dissertation Year Fellowship
University of California, Los Angeles Sept 2022 – June 2023
Student Travel Award
NeurIPS 2019, 2022, ICML 2018
Silver Medal Jan 2013
29th Chinese Mathematical Olympiad

TALKS

"Toward Generalizable Agents: Theory, Memory, and Multi-Agent Learning", July 2025
Adobe Research
"Variance-Dependent Regret Bounds for Non-stationary Linear Bandits", July 2024
25th International Symposium on Mathematical Programming
"Uncertainty-Aware Reward-Free Exploration with General Function Approximation", June 2024
The second RL theory workshop, University of Alberta
"Efficient Reinforcement Learning through Uncertainties", Feb-March 2023
School of Data Science, Chinese University of Hong Kong, Shenzhen
Department of Computer Science, University of California, Merced
Department of Computer Science, Chinese University of Hong Kong
Department of Industrial and Enterprise Systems Engineering, University department in Urbana, Illinois
Department of Computer Science, Indiana University Bloomington
Department of Computer Science and Engineering, University at Buffalo
Department of Electrical and Computer Engineering, Stony Brook University
Department of Computer Science, University of Rochester
Department of Computer Science, College of William & Mary
"Towards Efficient Reinforcement Learning", Dec 2022
Department of Computer Science and Engineering, Hong Kong University of Science and Technology
Department of Statistics, University of Florida
"Reinforcement Learning on Large State and Action Spaces", Nov 2022
LinkedIn AI seminar
"High-order Variance-reduced Gradients for Finite-sum Optimization", Oct 2022
INFORMS 2022
"Nearly Minimax Optimal Reinforcement Learning for Linear Mixture MDPs", Sep 2021
RL theory seminar
"Provable Efficient Algorithms for Neural Bandits", Aug 2021
EE&CS Department, University of California, Berkeley

PROFESSIONAL SERVICES

Conference Reviewer
Annual Conference on Neural Information Processing Systems (NeurIPS) 2019-2024
International Conference on Machine Learning (ICML) 2019-2024
International Conference on Learning Representations (ICLR) 2020-2023
International Joint Conferences on Artificial Intelligence (IJCAI) 2020, 2021
International Conference on Artificial Intelligence and Statistics (AISTATS) 2021-2023
Association for the Advancement of Artificial Intelligence (AAAI) 2020-2022
Journal Reviewer
Information and Inference: A Journal of the IMA
Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
Machine Learning (ML)
Journal of Machine Learning Research (JMLR)
Editorial Board Member
Journal of Artificial Intelligence (AIJ)
Grant Proposal Review
NSF SBIR, Panelist, 2024

Area Chair

International Conference on Machine Learning (ICML) 2025
International Conference on Learning Representations (ICLR) 2024

Research Experience

Research Intern
Manager: Aparna Pandey, Amazon AWS AI Jun 2022 - Sep 2022

- Self-supervised representation learning for time series anomaly detection

Research Intern
Mentor: Dr. Jing Huang, JD.COM Silicon Valley Research Center Jun 2021 - Sep 2021

- Improved the dialogue system with reinforcement learning methods (e.g., PG, TRPO) on neural language model (e.g., GPT-2)

Skills

Tools and Languages Python, LaTeX, R, MATLAB